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1  **
2  *College Closure Analysis
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4      *Labor Economics-USF 2025*
5  *Dr.Bobby Chung
6
7
8  * Set working directory
9  cd "C:\Users\juanc\OneDrive\Documents\College_Closure_Project\"
10
11 * Create folders for output
12 cap mkdir "output"
13 cap mkdir "output\tables"
14 cap mkdir "output\figures"
15
16
17 *LOAD AND CLEAN IPUMS ACS DATA
18
19 use "ipums_acs_2009_2019.dta", clear
20
21 * Keep only working-age population (16-64)
22 keep if age >= 16 & age <= 64
23
24 * Create 5-digit county FIPS code (combining state and county)
25 * First, check if we need to create it or if it already exists
26 capture confirm variable countyfip
27 if _rc == 0 {
28     * countyfip exists - check if it's already 5-digit or needs to be
    created
29     sum countyfip
30     if r(max) < 1000 {
31         * It's just the county portion, need to add state
32         replace countyfip = statefip*1000 + countyfip
33     }
34     * If max >= 1000, it's already the full 5-digit code, keep as is
35 }
36 else {
37     * countyfip doesn't exist, create it
38     gen countyfip = statefip*1000 + countyfip
39 }
40 label variable countyfip "5-digit County FIPS code"
41
42 * Keep only observations with valid county codes
43 drop if countyfip == . | countyfip == 0
44
45
46 *CREATE KEY VARIABLES
47
```

```
48 * Employment status variables
49 gen employed = (empstat == 1)
50 label variable employed "Employed"
51
52 gen unemployed = (empstat == 2)
53 label variable unemployed "Unemployed"
54
55 gen in_laborforce = (labforce == 2)
56 label variable in_laborforce "In labor force"
57
58 * Education sector workers (IND codes 7860–7890 = Educational
    services)
59 gen educ_sector = (ind >= 7860 & ind <= 7890 & employed == 1)
60 label variable educ_sector "Works in education sector"
61
62 * Wage variable (set N/A codes to missing)
63 replace incwage = . if incwage >= 999998
64 gen log_wage = ln(incwage) if incwage > 0
65 label variable log_wage "Log wage income"
66
67 * Demographic variables for controls
68 gen female = (sex == 2)
69 gen white = (race == 1)
70 gen black = (race == 2)
71 gen asian = (race == 4 | race == 5 | race == 6)
72 gen native = (race == 3)
73 gen hispanic = (hispan > 0 & hispan < 9)
74
75 * Education levels
76 gen hs_or_less = (educ <= 6)
77 gen some_college = (educ >= 7 & educ <= 9)
78 gen bachelors_plus = (educ >= 10)
79
80
81 * BASELINE EXPOSURE MEASURE (2009 ACS = 2005–2009)
82
83 preserve
84
85 * Keep only 2009 5-year sample (baseline)
86 keep if year == 2009
87
88 * Calculate total employment and education employment by county
89 collapse (sum) total_emp=employed ///
90             educ_emp=educ_sector ///
91             [pw=perwt], by(countyfip)
92
93 * Calculate exposure measure
94 gen exposure = educ_emp / total_emp
```

```
95 label variable exposure "Education employment share (2009 baseline)"
96
97 * Replace missing exposure with 0 (counties with no education
  employment)
98 replace exposure = 0 if exposure == .
99
100 * Save baseline exposure
101 save "baseline_exposure.dta", replace
102
103 restore
104
105 *CALCULATE PANEL OUTCOME VARIABLES (2010–2019)
106
107
108 * Keep only 2010–2019 data for outcomes
109 keep if year >= 2010 & year <= 2019
110
111 * Collapse to county-year level with all outcomes and controls
112 collapse (mean) employment_rate=employed ///
113                unemployment_rate=unemployed ///
114                lfp_rate=in_laborforce ///
115                median_wage=incwage ///
116                mean_log_wage=log_wage ///
117                pct_female=female ///
118                pct_white=white ///
119                pct_black=black ///
120                pct_asian=asian ///
121                pct_native=native ///
122                pct_hispanic=hispanic ///
123                pct_hs_or_less=hs_or_less ///
124                pct_some_college=some_college ///
125                pct_bachelors_plus=bachelors_plus ///
126                median_age=age ///
127                (count) population=perwt ///
128                [pw=perwt], by(countyfip year)
129
130 * Convert rates to percentages
131 replace employment_rate = employment_rate * 100
132 replace unemployment_rate = unemployment_rate * 100
133 replace lfp_rate = lfp_rate * 100
134 replace pct_female = pct_female * 100
135 replace pct_white = pct_white * 100
136 replace pct_black = pct_black * 100
137 replace pct_asian = pct_asian * 100
138 replace pct_hispanic = pct_hispanic * 100
139 replace pct_hs_or_less = pct_hs_or_less * 100
140 replace pct_some_college = pct_some_college * 100
141 replace pct_bachelors_plus = pct_bachelors_plus * 100
```

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142
143 * Label variables
144 label variable employment_rate "Employment rate (%)"
145 label variable unemployment_rate "Unemployment rate (%)"
146 label variable lfp_rate "Labor force participation rate (%)"
147 label variable median_wage "Median wage income ($)"
148 label variable mean_log_wage "Mean log wage"
149 label variable population "County population"
150
151 * Save panel data
152 save "panel_outcomes.dta", replace
153
154 *MERGE WITH CLOSURE DATA
155
156 * Load closure data
157 use "college_closures.dta", clear
158
159 * Keep only closures from 2010–2019 (drop always-treated)
160 keep if earliestyear >= 2010 & earliestyear <= 2019
161
162 * Keep only counties with non-missing enrollment data
163 * (This filters out counties with missing exposure data)
164 keep if eftytotlt != . & eaptot != .
165
166 * Save clean closure data
167 save "closure_clean.dta", replace
168
169 * Create full panel (all treated counties × all years 2010–2019)
170 expand 10
171 bysort countyfip: gen year = 2009 + _n
172 keep countyfip year earliestyear treat alwaystreat
173
174 * Create treatment variables
175 gen post = (year >= earliestyear)
176 label variable post "Post-closure period"
177
178 * Save closure panel
179 save "closure_panel.dta", replace
180
181 * Merge with outcome data
182 merge 1:1 countyfip year using "panel_outcomes.dta"
183 keep if _merge == 3 // Keep only matched observations
184 drop _merge
185
186 * Merge with baseline exposure
187 merge m:1 countyfip using "baseline_exposure.dta"
188 keep if _merge == 3 // Keep only counties with exposure data
189 drop _merge
```

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190
191 * Create interaction term
192 gen exposure_x_post = exposure * post
193 label variable exposure_x_post "Exposure × Post-closure"
194
195 * Save final analysis dataset
196 save "analysis_data.dta", replace
197
198
199 *SUMMARY STATISTICS
200
201
202 use "analysis_data.dta", clear
203
204 * Table 1: Summary statistics by treatment status
205 eststo clear
206
207 * Pre-period statistics (year < earliestyear)
208 eststo pre: estpost summarize employment_rate unemployment_rate ///
209     lfp_rate median_wage population exposure ///
210     if post == 0, detail
211
212 * Post-period statistics (year >= earliestyear)
213 eststo post: estpost summarize employment_rate unemployment_rate ///
214     lfp_rate median_wage population exposure ///
215     if post == 1, detail
216
217 * Export to LaTeX
218 esttab pre post using "output/tables/table1_summary_stats.tex", ///
219     cells("mean(fmt(2)) sd(fmt(2)) p25(fmt(2)) p50(fmt(2))
220     p75(fmt(2))") ///
221     label nomtitle nonumber replace booktabs ///
222     title("Summary Statistics by Treatment Period")
223
224 * Export to Excel
225 esttab pre post using "output/tables/table1_summary_stats.xlsx", ///
226     cells("mean sd min max") label replace
227
228 *MAIN DIFFERENCE-IN-DIFFERENCES REGRESSIONS
229
230
231 use "analysis_data.dta", clear
232
233 * Encode county for fixed effects
234 * countyfip is numeric, so use egen group instead of encode
235 egen county_id = group(countyfip)
236 label variable county_id "County ID for FE"

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237
238 eststo clear
239
240
241 *Employment Rate Outcomes
242
243 * Column 1: Basic DiD (no exposure interaction)
244 eststo emp1: reghdfe employment_rate post ///
245     pct_female pct_black pct_asian pct_hispanic ///
246     median_age pct_bachelors_plus, ///
247     absorb(county_id year) vce(cluster county_id)
248
249 * Column 2: DiD with exposure interaction (MAIN SPECIFICATION)
250 eststo emp2: reghdfe employment_rate post exposure_x_post ///
251     pct_female pct_black pct_asian pct_hispanic ///
252     median_age pct_bachelors_plus, ///
253     absorb(county_id year) vce(cluster county_id)
254
255 * Column 3: Unemployment rate – Basic DiD
256 eststo unemp1: reghdfe unemployment_rate post ///
257     pct_female pct_black pct_asian pct_hispanic ///
258     median_age pct_bachelors_plus, ///
259     absorb(county_id year) vce(cluster county_id)
260
261 * Column 4: Unemployment rate – with exposure
262 eststo unemp2: reghdfe unemployment_rate post exposure_x_post ///
263     pct_female pct_black pct_asian pct_hispanic ///
264     median_age pct_bachelors_plus, ///
265     absorb(county_id year) vce(cluster county_id)
266
267 * Export Table 2: Main Results
268 esttab emp1 emp2 unemp1 unemp2 using
269     "output/tables/table2_main_results.tex", ///
270     b(3) se(3) star(* 0.10 ** 0.05 *** 0.01) ///
271     label replace booktabs ///
272     mtitles("Emp Rate" "Emp Rate" "Unemp Rate" "Unemp Rate") ///
273     title("Main Results: Effect of College Closures on Labor Market
274     Outcomes") ///
275     addnotes("Standard errors clustered at county level in
276     parentheses." ///
277     "All regressions include county and year fixed effects."
278     ///
279     "Controls: % female, % black, % asian, % hispanic,
280     median age, % bachelors+")
281
282 esttab emp1 emp2 unemp1 unemp2 using
283     "output/tables/table2_main_results.xlsx", ///
284     b(3) se(3) star(* 0.10 ** 0.05 *** 0.01) label replace

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279
280
281 * Panel B: Wage and Population Outcomes
282
283
284 eststo clear
285
286 * Column 1: Median wages – Basic DiD
287 eststo wage1: reghdfe median_wage post ///
288     pct_female pct_black pct_asian pct_hispanic ///
289     median_age pct_bachelors_plus, ///
290     absorb(county_id year) vce(cluster county_id)
291
292 * Column 2: Median wages – with exposure
293 eststo wage2: reghdfe median_wage post exposure_x_post ///
294     pct_female pct_black pct_asian pct_hispanic ///
295     median_age pct_bachelors_plus, ///
296     absorb(county_id year) vce(cluster county_id)
297
298 * Column 3: Log wages – Basic DiD
299 eststo logwage1: reghdfe mean_log_wage post ///
300     pct_female pct_black pct_asian pct_hispanic ///
301     median_age pct_bachelors_plus, ///
302     absorb(county_id year) vce(cluster county_id)
303
304 * Column 4: Log wages – with exposure
305 eststo logwage2: reghdfe mean_log_wage post exposure_x_post ///
306     pct_female pct_black pct_asian pct_hispanic ///
307     median_age pct_bachelors_plus, ///
308     absorb(county_id year) vce(cluster county_id)
309
310 * Export Table 3: Wage Results
311 esttab wage1 wage2 logwage1 logwage2 using
312     "output/tables/table3_wage_results.tex", ///
313     b(3) se(3) star(* 0.10 ** 0.05 *** 0.01) ///
314     label replace booktabs ///
315     mtitles("Wage" "Wage" "Log Wage" "Log Wage") ///
316     title("Effect of College Closures on Wage Outcomes") ///
317     addnotes("Standard errors clustered at county level in
318     parentheses." ///
319     "All regressions include county and year fixed effects.")
320
321 esttab wage1 wage2 logwage1 logwage2 using
322     "output/tables/table3_wage_results.xlsx", ///
323     b(3) se(3) star(* 0.10 ** 0.05 *** 0.01) label replace
324
325 use "analysis_data.dta", clear
326
327

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323
324 * Recreate county_id for fixed effects
325 egen county_id = group(countyfip)
326
327 * Create exposure quartiles
328 xtile exposure_quartile = exposure if year == 2010, nq(4)
329 bysort countyfip: egen exposure_q = max(exposure_quartile)
330
331 * Generate quartile indicators
332 forvalues q = 1/4 {
333     gen exp_q`q' = (exposure_q == `q')
334     gen exp_q`q'_post = exp_q`q' * post
335 }
336
337 eststo clear
338
339 * Regression with quartile interactions
340 eststo hetero: reghdfe employment_rate post ///
341     exp_q2_post exp_q3_post exp_q4_post ///
342     pct_female pct_black pct_asian pct_hispanic ///
343     median_age pct_bachelors_plus, ///
344     absorb(county_id year) vce(cluster county_id)
345
346 esttab hetero using "output/tables/table4_heterogeneity.tex", ///
347     b(3) se(3) star(* 0.10 ** 0.05 *** 0.01) ///
348     label replace booktabs ///
349     title("Heterogeneous Effects by Education Sector Exposure") ///
350     addnotes("Quartile 1 (lowest exposure) is the omitted category."
351     ///
352     "Standard errors clustered at county level in
353     parentheses.")
354 *Add Figures**
355 use "analysis_data.dta", clear
356 egen county_id = group(countyfip)
357 * Figure 1: Distribution of closures over time
358 preserve
359 collapse (count) n_closures=countyfip, by(earliestyear)
360 twoway bar n_closures earliestyear, ///
361     xlabel(2010(1)2019) ///
362     ylabel(0 1000 2000) ///
363     ytitle("Number of Counties with Closures") ///
364     xtitle("Year of Closure") ///
365     title("College Closures by Year (2010–2019)") ///
366     graphregion(color(white)) bgcolor(white)
367 graph export "output/figures/figure1_closures_by_year.png", replace
368 width(3000)
369 graph export "output/figures/figure1_closures_by_year.pdf", replace
370 restore
371

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368 * Figure 2: Event study plot (employment rate)
369 preserve
370
371 * Create event time (years relative to closure)
372 gen event_time = year - earliestyear
373
374 * Keep only -5 to +5 years around closure
375 keep if event_time >= -5 & event_time <= 5
376
377 * Create shifted event time (add 6 so values are 1-11 instead of -5
  to 5)
378 gen event_time_shifted = event_time + 6
379 label define event_lbl 1 "-5" 2 "-4" 3 "-3" 4 "-2" 5 "-1" 6 "0" 7 "1"
  8 "2" 9 "3" 10 "4" 11 "5"
380 label values event_time_shifted event_lbl
381
382 * Run event study regression (omit -1, which is now 5 in shifted
  variable)
383 reghdfe employment_rate ib5.event_time_shifted ///
384     pct_female pct_black pct_asian pct_hispanic ///
385     median_age pct_bachelors_plus, ///
386     absorb(county_id year) vce(cluster county_id)
387
388 * Create event study plot
389 coefplot, vertical ///
390     drop(_cons pct_*) ///
391     yline(0, lcolor(red) lpattern(dash)) ///
392     xline(0, lcolor(gray) lpattern(dash)) ///
393     xtitle("Years Relative to Closure") ///
394     ytitle("Effect on Employment Rate (pp)") ///
395     title("Event Study: Employment Rate") ///
396     graphregion(color(white)) bgcolor(white)
397 graph export "output/figures/figure2_event_study.png", replace width(
  3000)
398 graph export "output/figures/figure2_event_study.pdf", replace
399
400 restore
401
402 * Figure 3: Exposure distribution
403 histogram exposure, ///
404     bin(30) ///
405     fraction ///
406     xtitle("Education Employment Share (2009)") ///
407     ytitle("Fraction of Counties") ///
408     title("Distribution of Education Sector Exposure") ///
409     graphregion(color(white)) bgcolor(white)
410 graph export "output/figures/figure3_exposure_dist.png", replace width
  (3000)

```

```

411 graph export "output/figures/figure3_exposure_dist.pdf", replace
412
413 * Figure 4: Treatment effect by exposure level
414 preserve
415
416 * Create exposure deciles
417 xtile exposure_decile = exposure if year == 2010, nq(10)
418 bysort countyfip: egen exp_dec = max(exposure_decile)
419
420 * Calculate mean exposure by decile
421 bysort exp_dec: egen mean_exposure = mean(exposure)
422
423 * Run regression for each decile
424 gen effect = .
425 gen ci_low = .
426 gen ci_high = .
427
428 forvalues d = 1/10 {
429     qui reghdfe employment_rate post if exp_dec == `d', ///
430         absorb(county_id year) vce(cluster county_id)
431     replace effect = _b[post] if exp_dec == `d'
432     replace ci_low = _b[post] - 1.96*_se[post] if exp_dec == `d'
433     replace ci_high = _b[post] + 1.96*_se[post] if exp_dec == `d'
434 }
435
436 * Collapse to decile level
437 collapse (mean) mean_exposure effect ci_low ci_high, by(exp_dec)
438
439 * Plot
440 twoway (rarea ci_low ci_high mean_exposure, color(gs14)) ///
441     (line effect mean_exposure, lcolor(navy) lwidth(thick)), ///
442     yline(0, lcolor(red) lpattern(dash)) ///
443     xtitle("Education Employment Share") ///
444     ytitle("Effect on Employment Rate (pp)") ///
445     title("Treatment Effect Heterogeneity by Exposure") ///
446     legend(order(2 "Point Estimate" 1 "95% CI")) ///
447     graphregion(color(white)) bgcolor(white)
448 graph export "output/figures/figure4_effect_by_exposure.png", replace
449     width(3000)
450 graph export "output/figures/figure4_effect_by_exposure.pdf", replace
451
452 restore
453
454 ***ROBUSTNESS CHECKS***
455
456
457 use "analysis_data.dta", clear

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```
458
459 * Recreate county_id for fixed effects
460 egen county_id = group(countyfip)
461
462 eststo clear
463
464 * Robustness 1: Different control sets
465 eststo robust1: reghdfe employment_rate post exposure_x_post, ///
466     absorb(county_id year) vce(cluster county_id)
467
468 eststo robust2: reghdfe employment_rate post exposure_x_post ///
469     pct_female median_age, ///
470     absorb(county_id year) vce(cluster county_id)
471
472 eststo robust3: reghdfe employment_rate post exposure_x_post ///
473     pct_female pct_black pct_asian pct_hispanic ///
474     median_age pct_bachelors_plus pct_some_college, ///
475     absorb(county_id year) vce(cluster county_id)
476
477 * Export robustness table
478 esttab robust1 robust2 robust3 using
479     "output/tables/table5_robustness.tex", ///
480     b(3) se(3) star(* 0.10 ** 0.05 *** 0.01) ///
481     label replace booktabs ///
482     mtitles("No Controls" "Basic Controls" "Full Controls") ///
483     title("Robustness: Different Control Specifications")
```